

Indices, logs and graphing
common functions

Key topics

Graphing common functions

Using and combining indices

Using and combining logs

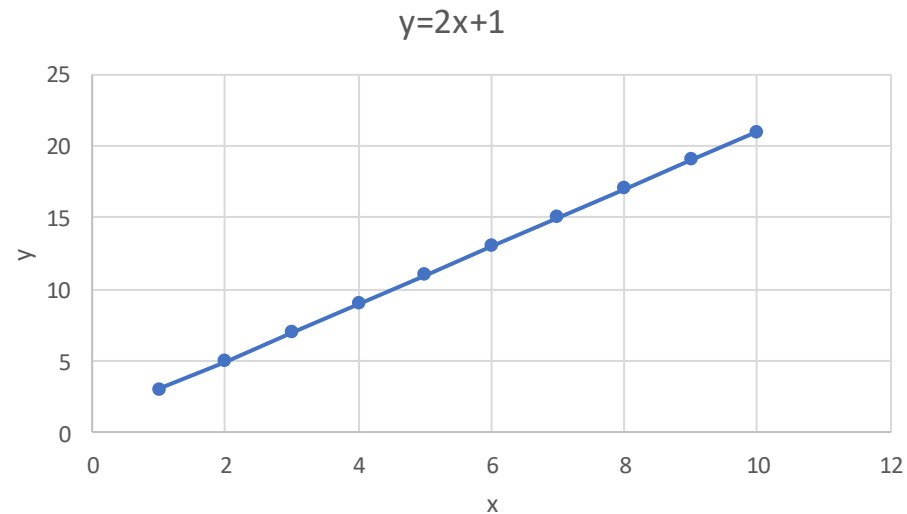
Dominance in graphs

$$y=mx+c$$

This is the equation of a line (curve!) and is a common function to use
e.g. $y = 2x + 1$

$$m = 2, c = 1$$

x	mx	c	y
1	2	1	3
2	4	1	5
3	6	1	7
4	8	1	9
5	10	1	11
6	12	1	13
7	14	1	15
8	16	1	17
9	18	1	19
10	20	1	21

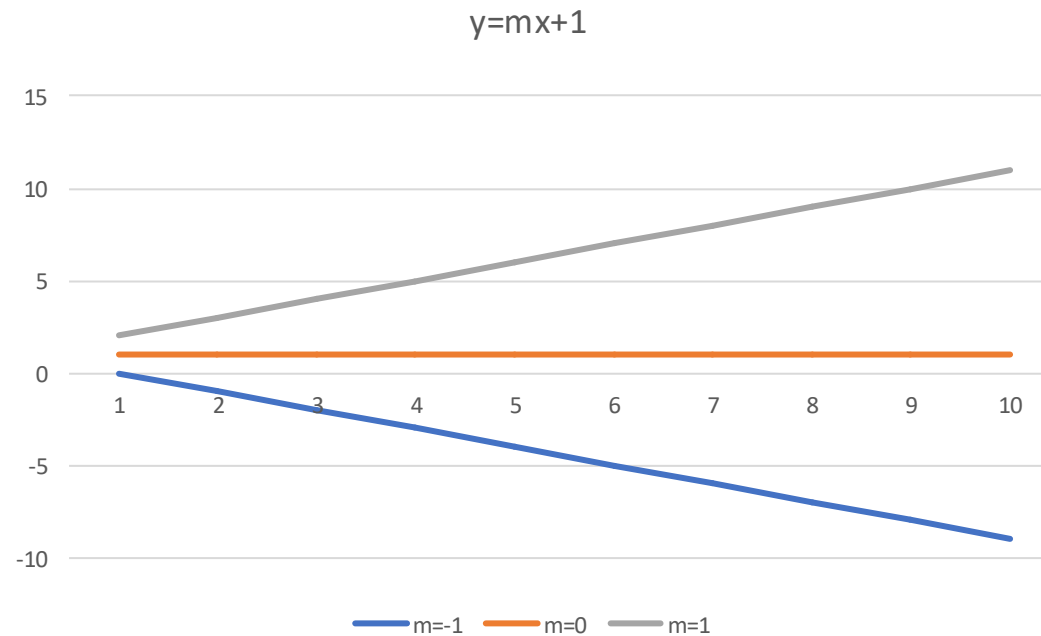
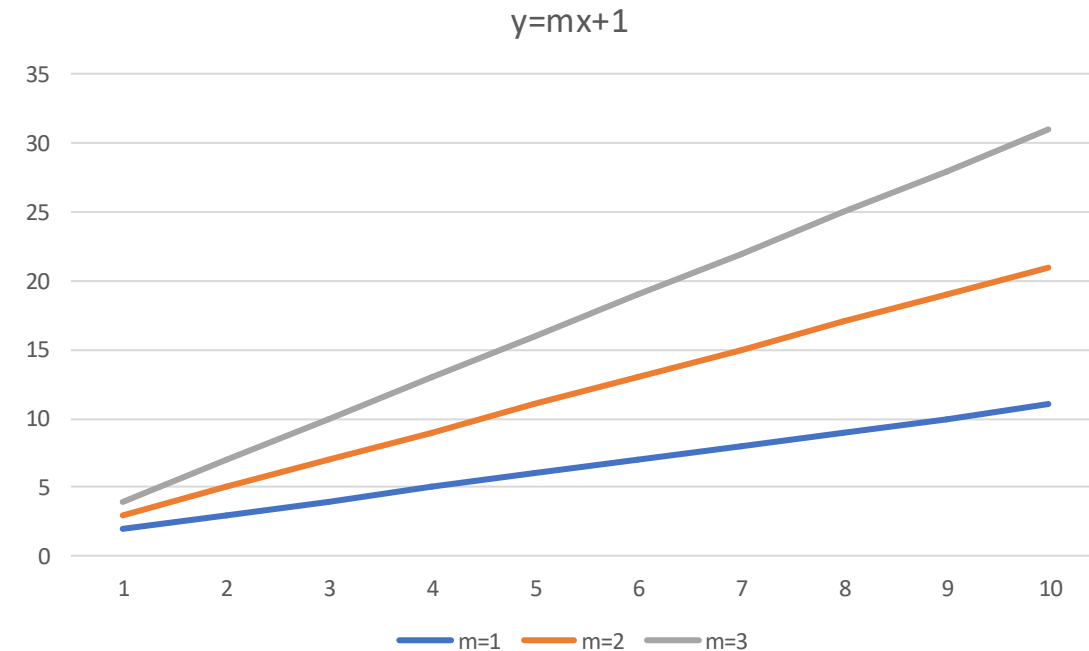


m – the slope

The greater m , the greater the slope of the line

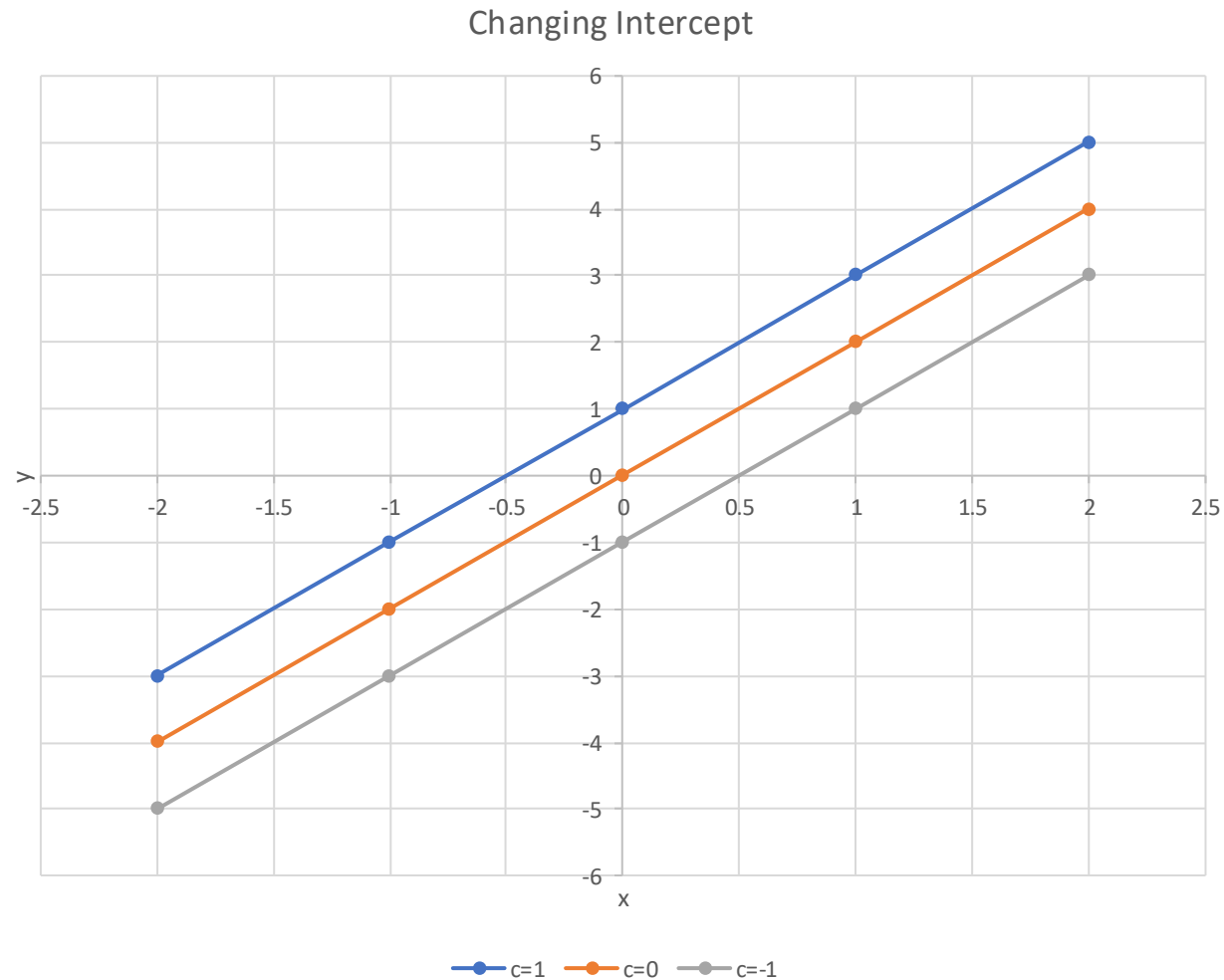
m can be negative, again, the larger m is the steeper the slope

m can take the value zero also, in this case $y = c$



C – the intercept

The point where the line crosses the y axis
(where $x = 0, y = c$)



Power terminology

Repeated addition of a number is called **multiplication**:

$3+3+3+3$ is four additions of three which is $4 \times 3 = 12$

Repeated multiplication of a number is called **exponentiation**:

$$3 \times 3 \times 3 \times 3 = 3^4 = 81$$

When we write a^n a is the **base** and n is the **exponent**

Power notation

x^2 could be written x^2 , especially if people don't have access to tools like MSE.

$$x \times x = x^2$$

Integer powers are the most common

$$\frac{1}{x} = x^{-1}$$

Negative powers are also possible

$$\sqrt{x} = x^{\frac{1}{2}}$$

Fractional powers exist too!

Standard form

Examples:

Generally written in the form $a \times 10^n$

Where $1 \leq a \leq 10$ and both a and n can be positive or negative

A positive exponent means 'move the decimal point this many places to the right' and a negative one means to move to the left

$$0.0024 = 2.4 \times 10^{-3}$$

$$0.000\ 001 = 1 \times 10^{-6}$$

$$57 = 5.7 \times 10^1$$

$$1000\ 000 = 1 \times 10^6$$

$$5.7 = 5.7 \times 10^0$$

Powers as multiples

$$a^3 = a \times a \times a$$

$$a^2 = a \times a$$

$$a^1 = a$$

$$a^0 = \frac{a}{a} = 1$$

$$a^{-1} = \frac{1}{a}$$

$$a^{-2} = \frac{1}{a^2}$$

$$a^{-3} = \frac{1}{a^3}$$

Divide by a

It is helpful to think of powers as repeated multiplication.

It makes it easier to understand powers of zero or negative numbers

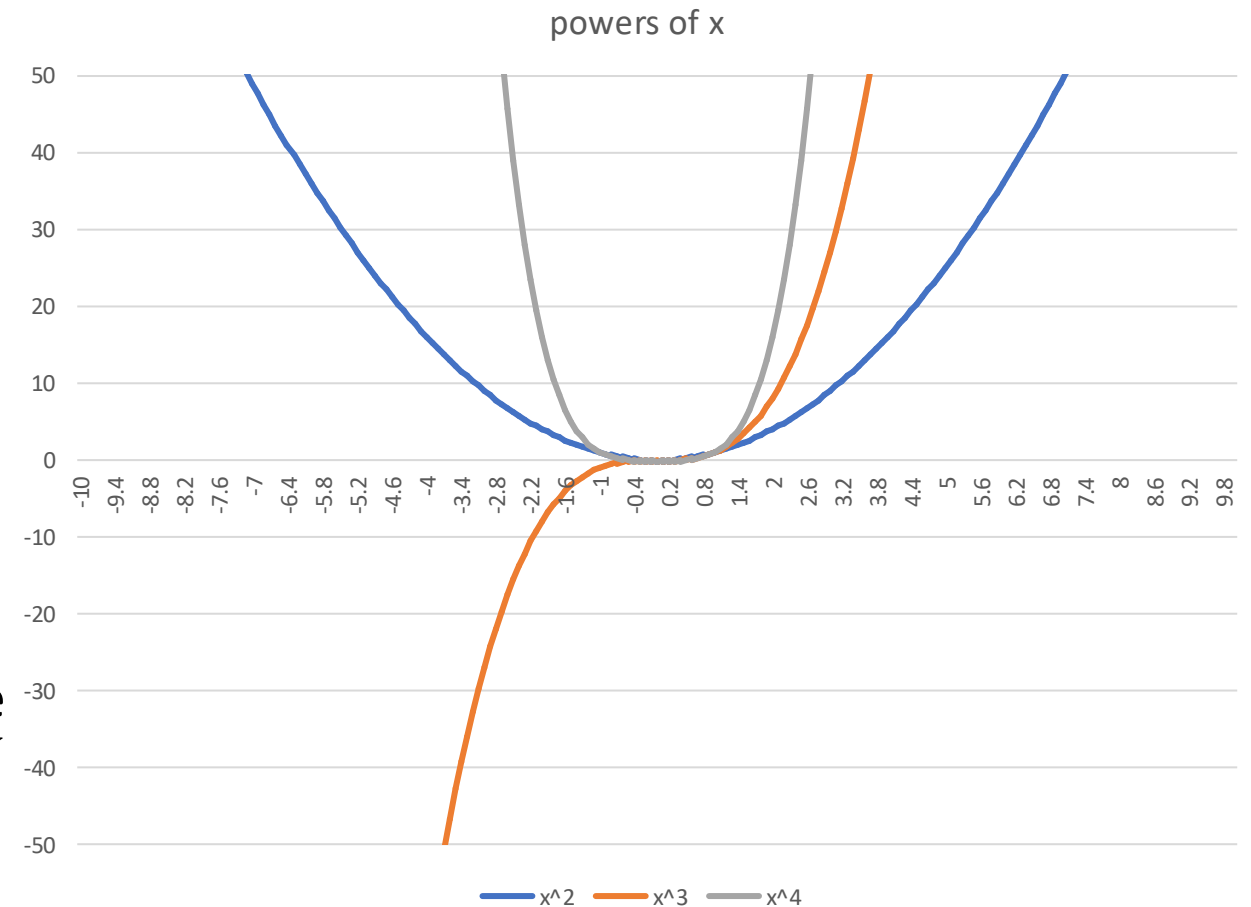
Graphing Powers of a variable

We have used x^2 and x^3 etc.

$$\begin{aligned}x \times x &= x^2 \\x \times x \times x &= x^3 \\x \times x \times x \times x &= x^4\end{aligned}$$

When we graph them, we can see that x^3 looks quite different to the others

It is an odd function, while even powers give even functions (are symmetrical about the y axis)



Even Powers

Even powers can only ever have positive results. Multiplying something by itself always gives a positive value, even if you start with a negative

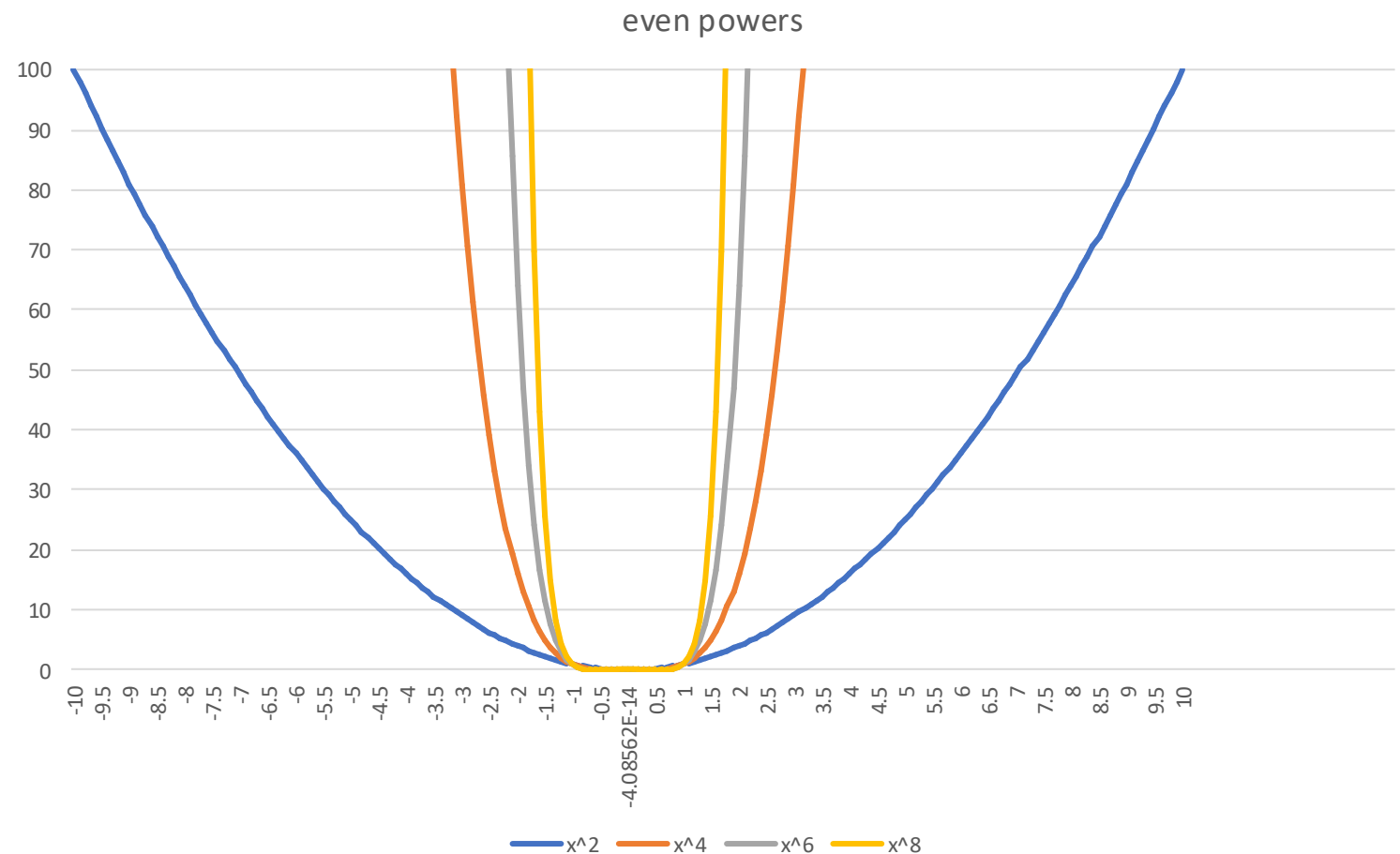
The higher the power, the steeper the slope

$$x \times x = x^2$$

$$x \times x \times x \times x = x^4$$

$$x \times x \times x \times x \times x \times x = x^6$$

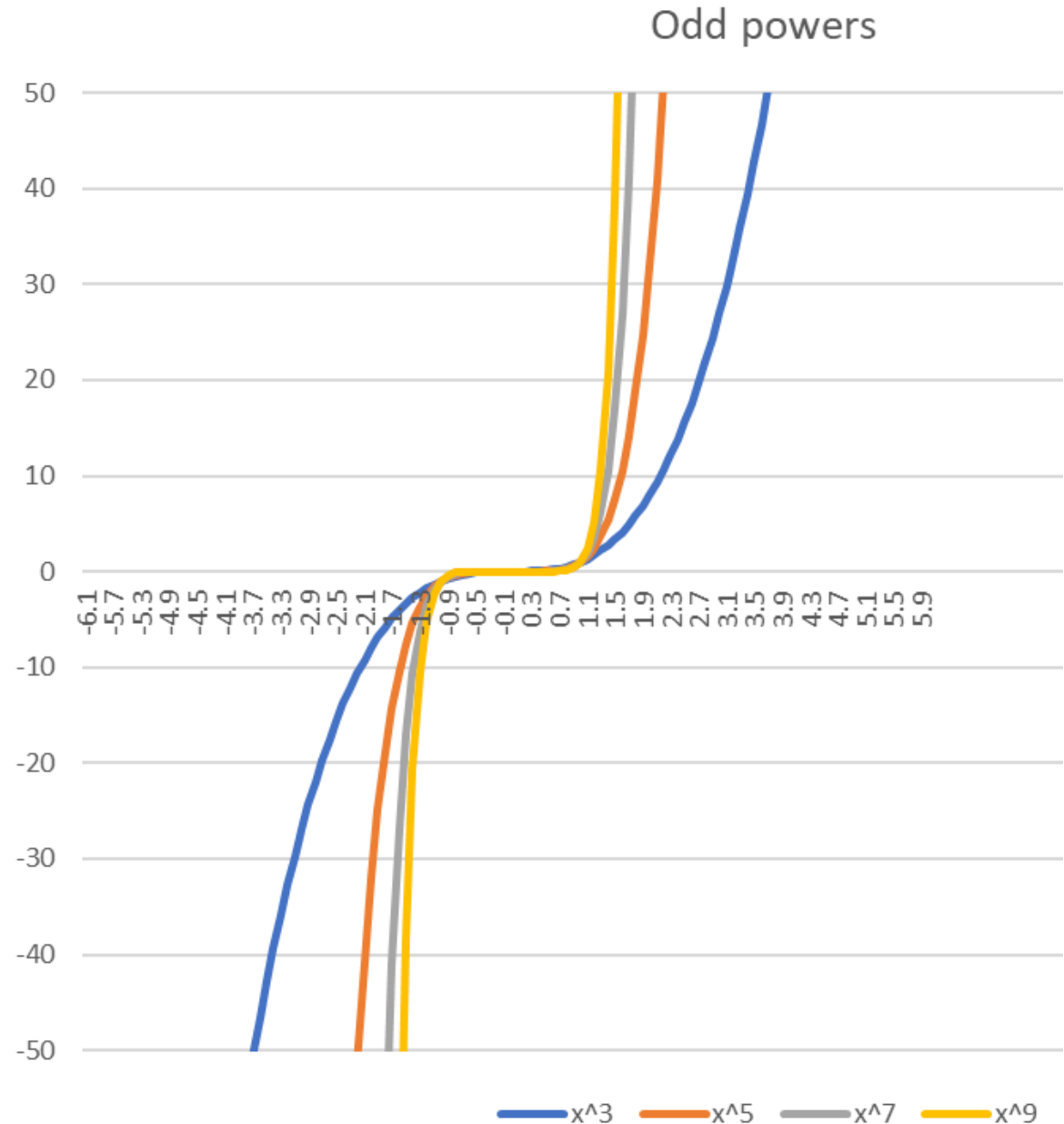
$$x \times x \times x \times x \times x \times x \times x \times x = x^8$$



Odd Powers

Odd powers take negative values below zero, positive above zero

The higher the power, the steeper the slope, on both sides of the axis



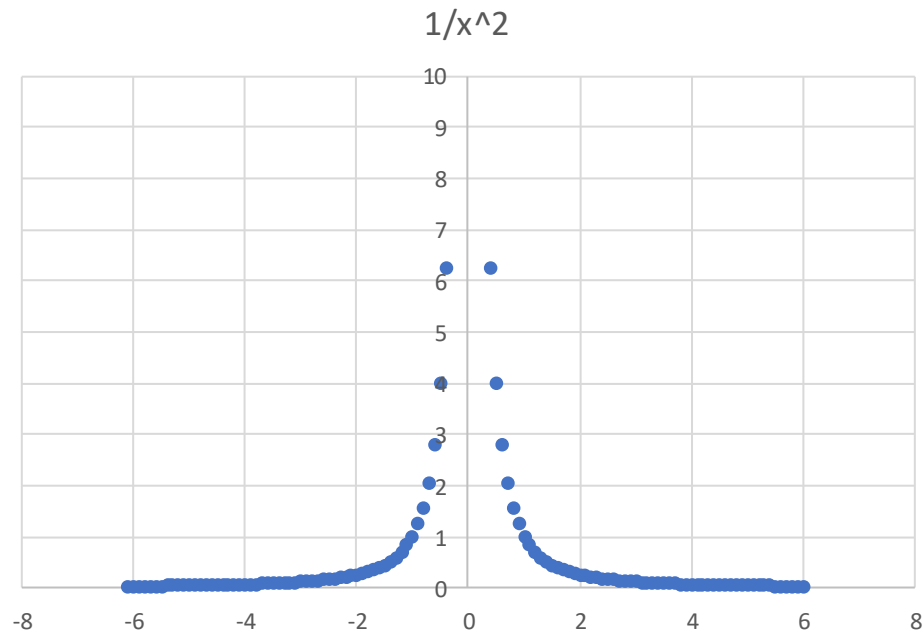
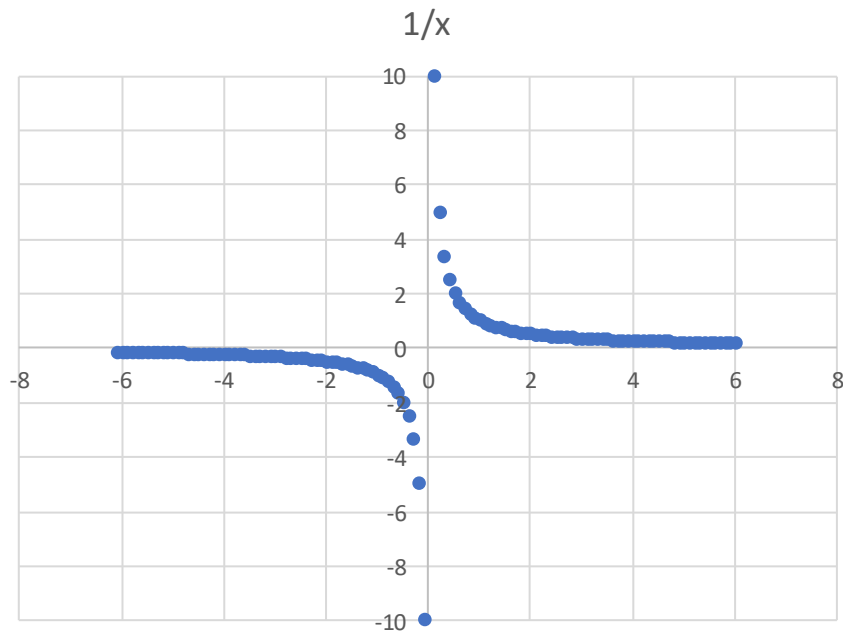
Negative Powers

Negative powers also change according to whether they are odd or even

$$\frac{1}{x} = x^{-1}$$

$$\frac{1}{x^2} = x^{-2}$$

$$\frac{1}{x^3} = x^{-3}$$



Combining powers (1)

$$a^m \times a^n = a^{m+n}$$

example 1: $x^2 \times x^3 = (x \times x)(x \times x \times x) = x^{2+3} = x^5$

example 2: $y \times y^7 = y^{1+7} = y^8$

$$\frac{a^m}{a^n} = a^{m-n}$$

example 1: $\frac{x^3}{x^2} = \frac{x \times x \times x}{x \times x} x^{3-2} = x^1 = x$

example 2: $\frac{x^3}{x^3} = x^{3-3} = x^0 = 1$

This only works when the two variables you are multiplying have the same base!

Combining powers (2)

$$(a^m)^n = a^{mn}$$

example: $(x^2)^3 = (x \times x)(x \times x)(x \times x) = x^{2 \times 3} = x^6$

Think about a special case of this:

$$(a^{\frac{1}{n}})^n = a^{n \frac{1}{n}} = a^{\frac{n}{n}} = a$$

Hence, $a^{\frac{1}{n}}$ has the property that, when raised to the nth power we get a.

So: $a^{\frac{1}{n}} = \sqrt[n]{a}$

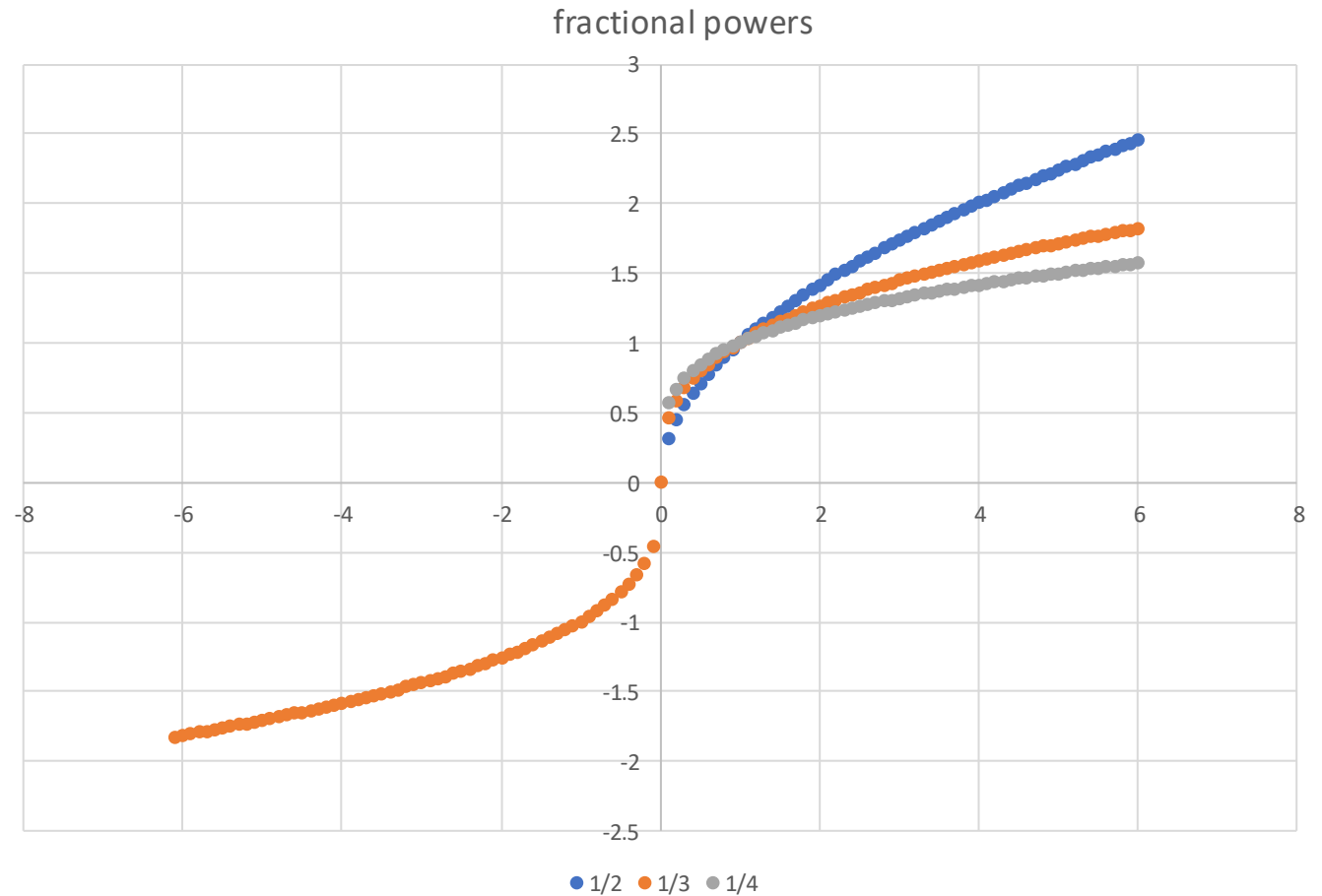
Fractional powers

Fractional powers behave strangely – the even fractions ($1/2$, $1/4$, $1/6$ etc.) only exist for positive values of x .

The odd fractions ($1/3$, $1/5$ etc.) exist for negative values as well

The magnitude of the fraction determines the shape of the curve differently below 1 and above 1!

$$\sqrt[2]{x} = x^{1/2}$$
$$\sqrt[3]{x} = x^{1/3}$$
$$\sqrt[4]{x} = x^{1/4}$$



Combining powers (3)

$$(ab)^n = a^n b^n$$

example: $(xy)^3 = xy \times xy \times xy = x^3 y^3$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

example: $\left(\frac{x}{y}\right)^3 = \frac{x}{y} \times \frac{x}{y} \times \frac{x}{y} = \frac{x^3}{y^3}$

Logarithms

Generally written in the form: $y = a^x$ where y is the **number**, a is the **base** and x is the **logarithm**

If $y = a^x$ then we write $x = \log_a y$ **“log to the base a of y”**

Examples:

$100 = 10^2$ so it follows that $2 = \log_{10} 100$

$8 = 2^3$ is the same as $3 = \log_2 8$

$16 = 4^2$ is the same as $2 = \log_4 16$

**“2 is the power I need to apply to 4,
to get an answer of 16”**

Using Logs in equations

We can apply logs to both sides of an equation, just like we might square both sides or divide both sides:

$$8 = 2^x \text{ is the same as } \log_{10} 8 = \log_{10} 2^x$$

In plain English: If Eight is equal to Two, raised to the power of x , then the power I have to raise Ten to, in order to get Eight is the same power I have to raise Ten to, in order to get Two, raised to the power of x .

In this case, $10^{0.90309} = 8$ so $10^{0.90309} = 2^x$

Laws of logs

$$\log a + \log b = \log ab$$

example: $\log_{10}(a^2) + \log_{10}(b^3) = \log_{10}(a^2 b^3)$

$$\log a - \log b = \log \frac{a}{b}$$

example: $\log_{10} \left(\frac{a^2}{b^3} \right) = \log_{10}(a^2) - \log_{10}(b^3)$

Laws of logs 2

$$\log a^n = n \log a$$

example: $\log_{10}(a^2) + \log_{10}(b^3) = \log_{10}(a^2 b^3)$

$$\log_a N = \frac{\log_b N}{\log_b a}$$

example: $\log_5 15 = \frac{\log_{10} 15}{\log_{10} 5} = \frac{1.1761}{0.6990} = 1.6825$

Question 1

Express $\log_{10} \left(\frac{a^2 b^3}{100\sqrt{c}} \right)$ as separate terms of logs of a , b and c :

$$\log_{10} \left(\frac{a^2 b^3}{100\sqrt{c}} \right) = \log_{10}(a^2 b^3) - \log_{10}(100\sqrt{c}) \quad \log a - \log b = \log \frac{a}{b}$$

$$= (\log_{10}(a^2) + \log_{10}(b^3)) - (\log_{10}(100) + \log_{10}(\sqrt{c})) \quad \log a + \log b = \log ab$$

$$= 2\log_{10}a + 3\log_{10}b - 2 - \frac{1}{2}\log_{10}c \quad \log a^n = n \log a$$

Natural log e

We will look at how e arises in another class

$$e = 2.71828 \dots = \left(1 + \frac{1}{n}\right)^n \text{ as } n \rightarrow \infty$$

- 1) Like Pi, nobody 'invented' e, it arises naturally when we look at exponential problems
- 2) Like Pi, it is **irrational**, its decimal places go on forever
- 3) Also termed the '**Naperian**' log, in some sources, to distinguish from the '**Briggsian**' log, which uses base 10
- 4) We will come back to e, for now just remember it is the number 2.718 and you'll be fine.
- 5) Rather than writing $\log_e$ we write $\ln$
- 6) Another written form of e^x is $\exp(x)$

Question 2

The time for a current to decrease from I to i is obtained from the formula

$$i = Ie^{-\frac{t}{CR}}$$

obtain an expression for the time, t

$$i = Ie^{-\frac{t}{CR}}$$

$$i = \frac{I}{e^{\frac{t}{CR}}}$$

$$e^{\frac{t}{CR}} = \frac{I}{i}$$

$$\frac{t}{CR} = \ln\left(\frac{I}{i}\right)$$

$$t = CR \ln\left(\frac{I}{i}\right)$$

Question 3a

In resisted motion, the velocity v at time t is given in terms of the initial velocity u by the formula

$$v = u(1 - e^{-kt})$$

obtain an expression for the time, t :

$$v = u(1 - e^{-kt})$$

$$\frac{v}{u} = 1 - e^{-kt}$$

$$\frac{v}{u} - 1 = -e^{-kt}$$

$$e^{-kt} = 1 - \frac{v}{u}$$

$$\frac{1}{e^{kt}} = 1 - \frac{v}{u}$$

$$\frac{1}{\left(1 - \frac{v}{u}\right)} = e^{kt}$$

$$\ln\left(\frac{1}{\left(1 - \frac{v}{u}\right)}\right) = kt$$

$$t = \frac{1}{k} \ln\left(\frac{1}{\left(1 - \frac{v}{u}\right)}\right) = \frac{1}{k} \ln\left(\frac{u}{\left(u - \frac{uv}{u}\right)}\right) = \frac{1}{k} \ln\left(\frac{u}{u - v}\right)$$

Question 3b

In resisted motion, the velocity v at time t is given in terms of the initial velocity u by the formula

$$v = u(1 - e^{-kt})$$

obtain an expression for the time, t :

$$v = u(1 - e^{-kt})$$

$$v = u - ue^{-kt}$$

$$v = u - ue^{-kt}$$

$$v = u - \frac{u}{e^{kt}}$$

$$ve^{kt} = ue^{kt} - u$$

$$u = (u - v)e^{kt}$$

$$\frac{u}{u - v} = e^{kt}$$

$$kt = \ln\left(\frac{u}{u - v}\right)$$

$$t = \frac{1}{k} \ln\left(\frac{u}{u - v}\right)$$

Logs Vs indices

Think about $y = x^2$ as x rises from zero. It gets large fast – every step the amount the graph rises increases:

From $x = 2$ to $x = 3$ y rises five points, from 4 to 9
from $x = 3$ to $x = 4$ y rises seven points, from 9 to 16

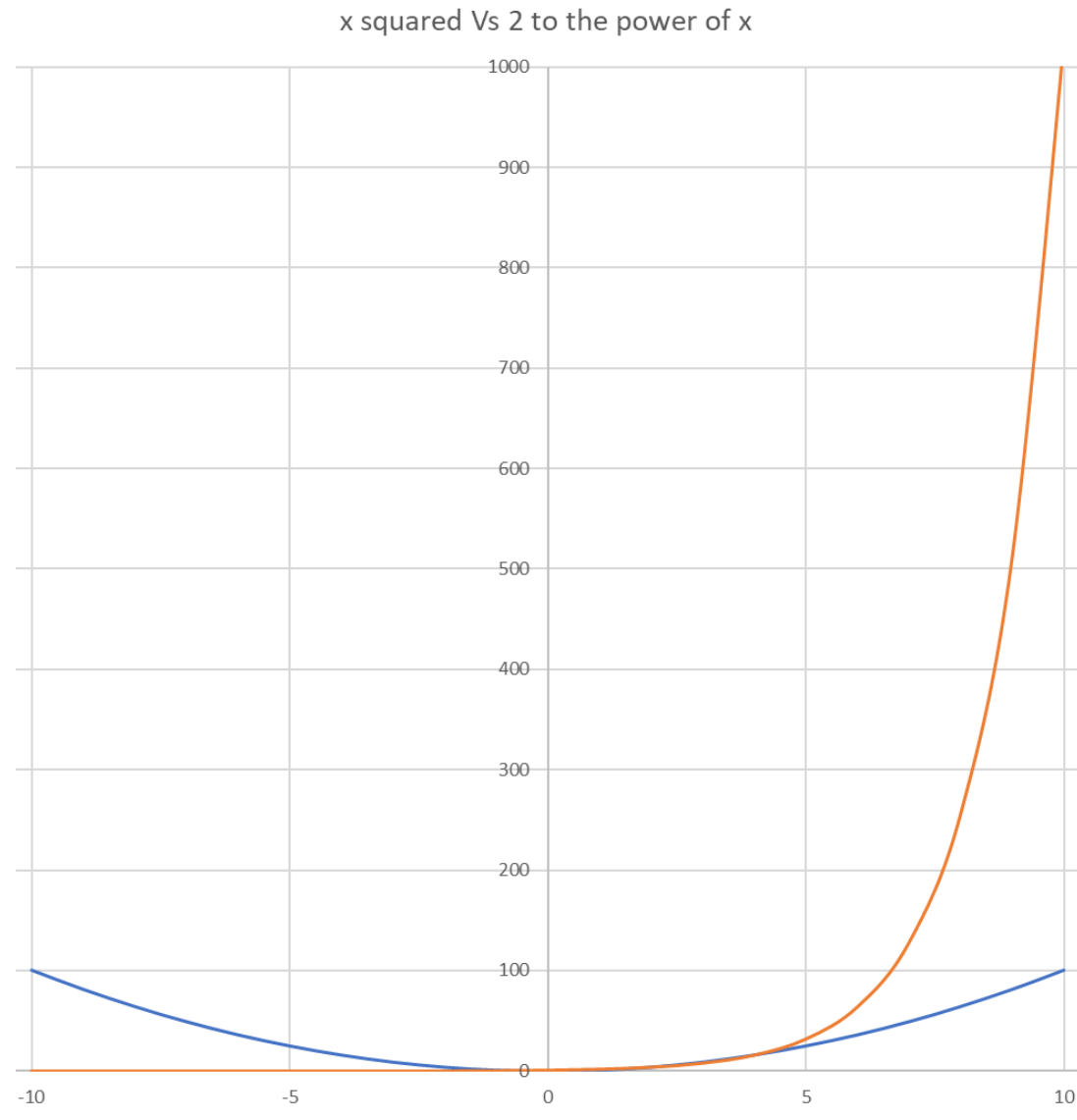
Now think about $y = 2^x$ as x rises from zero. When $x = 2$, y behaves similarly to the graph above, when $x = 3$, y rises even faster and when $x = 4$, y rises faster still!

The behaviour for negative values is extremely different -

This is what distinguishes an **exponential** relationship from a **power** relationship

Power Vs. Exponential

x	x^2	2^x
-10	100	0.000977
-9	81	0.001953
-8	64	0.003906
-7	49	0.007813
-6	36	0.015625
-5	25	0.03125
-4	16	0.0625
-3	9	0.125
-2	4	0.25
-1	1	0.5
0	0	1
1	1	2
2	4	4
3	9	8
4	16	16
5	25	32
6	36	64
7	49	128
8	64	256
9	81	512
10	100	1024

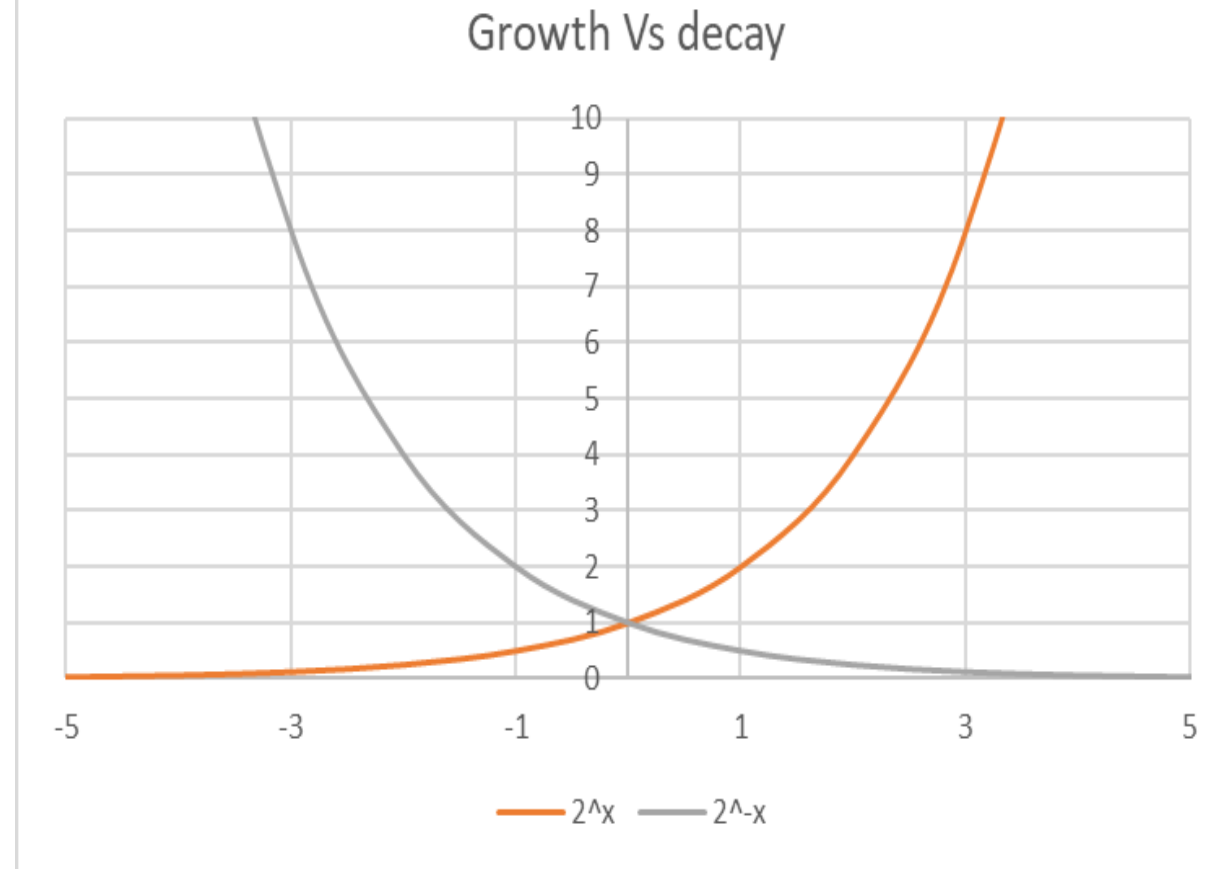


Growth Vs decay

Here we compare $y = 2^x$ with $y = 2^{-x}$

Growth used to model compound interest accruing on savings (or student debt!)

Decay used to model transient electric current, radioactive decay etc.



Dominance in graphs

Functions composed of multiple terms (polynomials) show a range of characteristics – often certain characteristics are dominant over certain ranges

Consider $y = ax^m + \dots + bx^n + c$

For **large** values of x it behaves like $y = ax^m$,
for **small** values of x it behaves like $y = bx^n + c$

Dominance example

$$y = x^3 - 2x$$

For large values of x it looks
like $y = x^3$

For small values of x it looks
like $y = -2x + 0$

